THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Assignment 4, 2025

School of Mathematics and Statistics

MAST20009 Vector Calculus

Submission deadline: 5pm on Monday May 26

This assignment consists of 11 pages (including this page) with 4 questions and 44 total marks

Instructions to Students

- This assignment is worth 5% of your total MAST20009 mark.
- If you have a printer, print the assignment one-sided.

Writing

- Assignments must be neatly written in blue or black pen or may be typeset using LaTeX.
- Full working must be shown in your solutions.
- Marks will be deducted for incomplete working, insufficient justification of steps, incorrect mathematical notation and for messy presentation of solutions.
- If you are writing answers on the assignment and you need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

Scanning and Submitting

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned assignment as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary.

Page 1 of 11 — add any extra pages after page 11 — Page 1 of 11

Page 2 of 11 — add any extra pages after page 11 — Page 2 of 11

Blank page

Page 3 of 11 — add any extra pages after page 11 — Page 3 of 11

Question 1 (8 marks)

Let $P = ABCD$ be the parallelogram in the xy-plane with

$$A = (0,0), \quad B = (-2,1), \quad C = (-1,4), \quad D = (1,3).$$

Let $P' = A'B'C'D'$ be the square in the uv-plane with

$$A' = (0,0), \quad B' = (1,0), \quad C' = (1,1), \quad D' = (0,1).$$

(a) Find a bijective linear map which takes P' to P, such that $A' \mapsto A$, $B' \mapsto B$, $C' \mapsto C$, and $D' \mapsto D$.

We can parametrise the points X in P by

$$\overrightarrow{OX} = \overrightarrow{OA} + u\overrightarrow{AB} + v\overrightarrow{AD}, \quad 0 \le u \le 1, 0 \le v \le 1.$$

Then $\overrightarrow{OA} = (0,0)$, $\overrightarrow{AB} = (-2,1)$ and $\overrightarrow{AD} = (1,3)$.

So if $X = (x,y)$, we have

$$x = -2u + v$$
$$y = u + 3v$$

Page 4 of 11 — add any extra pages after page 11 — Page 4 of 11

(b) Evaluate the double integral

$$\iint_P (xy + x^2)\, dx\, dy.$$

Use a change of variables. We need the Jacobian

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} -2 & 1 \\ 1 & 3 \end{vmatrix} = -7.$$

Then

$$\begin{aligned}
&\iint_P (xy + x^2)\, dx\, dy \\
&= \iint_{P'} ((-2u+v)(u+3v) + (-2u+v)^2)|-7|\, du\, dv \\
&= 7 \int_0^1 \int_0^1 (2u^2 - 9uv + 4v^2)\, du\, dv \\
&= 7 \int_0^1 \left(\frac{2}{3} - \frac{9v}{2} + 4v^2 \right) dv \\
&= -\frac{7}{4}
\end{aligned}$$

Page 5 of 11 — add any extra pages after page 11 — Page 5 of 11

Question 2 (10 marks)

Let S be the ellipsoid defined by

$$4x^2 + 4y^2 + z^2 = 4.$$

Write down a double integral which gives the surface area of that part of S which is above the xy-plane. The integrand should be an explicit scalar function, i.e. not written in terms of a cross product. *You do not need to evaluate the integral.*

To parametrise S we use ellipsoidal coordinates. Use $\mathbf{T}(\theta, \phi) = (x, y, z)$, with

$$\begin{aligned} x &= \cos\phi \sin\theta \\ y &= \sin\phi \sin\theta \\ z &= 2\cos\theta \end{aligned}$$

so that

$$4x^2 + 4y^2 + z^2 = 4\cos^2\phi \sin^2\theta + 4\sin^2\phi \sin^2\theta + 4\cos^2\theta = 4.$$

We have $0 \leq \phi \leq 2\pi$. For θ, if we must stay above the xy-plane then $0 \leq \theta \leq \frac{\pi}{2}$.

Next to integrate over the surface, we need $\mathbf{T}_\theta \times \mathbf{T}_\phi$, which is

$$\begin{aligned} \mathbf{T}_\theta \times \mathbf{T}_\phi &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos\phi\cos\theta & \sin\phi\cos\theta & -2\sin\theta \\ -\sin\phi\sin\theta & \cos\phi\sin\theta & 0 \end{vmatrix} \\ &= (2\cos\phi\sin^2\theta, 2\sin\phi\sin^2\theta, \cos\theta\sin\theta) \end{aligned}$$

Then

$$|\mathbf{T}_\theta \times \mathbf{T}_\phi| = \cdots = \sin\theta\sqrt{3\sin^2\theta + 1}.$$

The integral we want is then

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin\theta\sqrt{3\sin^2\theta + 1}\, d\theta\, d\phi.$$

This turns out to be equal to

$$= \frac{4\pi^2}{3\sqrt{3}} + \pi.$$

Page 6 of 11 — add any extra pages after page 11 — Page 6 of 11

Page 7 of 11 — add any extra pages after page 11 — Page 7 of 11

Question 3 (14 marks)

Let S_1 be the surface of the cone $z = \sqrt{x^2 + y^2}$, $0 \le z \le 1$, and let S_2 be the round "lid" at the top of S_1. Let $S = S_1 \cup S_2$, so that S is a closed surface. Orient S so that its normal vectors are in the outward direction.

(a) Find a parametrisation $\mathbf{T}$ of S_1 and write down its normal vector.

S_1 is a cone so cylindrical coordinates are appropriate. Set

$$x = r\cos\phi, \quad y = r\sin\phi, \quad z = \sqrt{x^2 + y^2} = r$$

with $0 \le \phi \le 2\pi$, $0 \le r \le 1$. Write

$$\mathbf{T}(r, \phi) = (x, y, z) = (r\cos\phi, r\sin\phi, r).$$

Then

$$\begin{aligned} \mathbf{T}_r &= (\cos\phi, \sin\phi, 1) \\ \mathbf{T}_\phi &= (-r\sin\phi, r\cos\phi, 0). \end{aligned}$$

For outward normals we need

$$\mathbf{T}_\phi \times \mathbf{T}_r = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -r\sin\phi & r\cos\phi & 0 \\ \cos\phi & \sin\phi & 1 \end{vmatrix} = (r\cos\phi, r\sin\phi, -r).$$

As a sanity check we can see that the $\mathbf{k}$ component is negative, i.e. this cross product points "downwards", which is what we expect.

Page 8 of 11 — add any extra pages after page 11 — Page 8 of 11

(b) Let $\mathbf{F}$ be the vector field $\mathbf{F}(x, y, z) = (y, 0, z)$. Compute the flux

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS,$$

where $\hat{\mathbf{n}}$ is the outward normal to the surface S.

We must deal with S_1 and S_2 separately. For S_2, the outward normal vector is just $\mathbf{k}$, so (recalling that on S_2 we have $z = 1$)

$$\begin{aligned}\iint_{S_2} \mathbf{F} \cdot \hat{\mathbf{n}}\, dS &= \iint_{S_2} (y, 0, z) \cdot (0, 0, 1)\, dS \\ &= \iint_{S_2} dS \\ &= (\text{area of } S_1) = \pi.\end{aligned}$$

Meanwhile on S_1,

$$\begin{aligned}\iint_{S_1} \mathbf{F} \cdot \hat{\mathbf{n}}\, dS &= \int_0^{2\pi} \int_0^1 (r \sin\phi, 0, r) \cdot (\mathbf{T}_\phi \times \mathbf{T}_r)\, dr\, d\phi \\ &= \int_0^{2\pi} \int_0^1 (r^2 \cos\phi \sin\phi - r^2)\, dr\, d\phi \\ &= \int_0^1 r^2\, dr \int_0^{2\pi} (\cos\phi \sin\phi - 1)\, d\phi \\ &= \frac{1}{3} \int_0^{2\pi} (\tfrac{1}{2} \sin 2\phi - 1)\, d\phi \\ &= \frac{1}{3} \left[-\frac{1}{4} \cos 2\phi - \phi \right]_{r=0}^{r=2\pi} = -\frac{2\pi}{3}\end{aligned}$$

Combining the two integrals gives

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS = \pi - \frac{2\pi}{3} = \frac{\pi}{3}.$$

Page 9 of 11 — add any extra pages after page 11 — Page 9 of 11

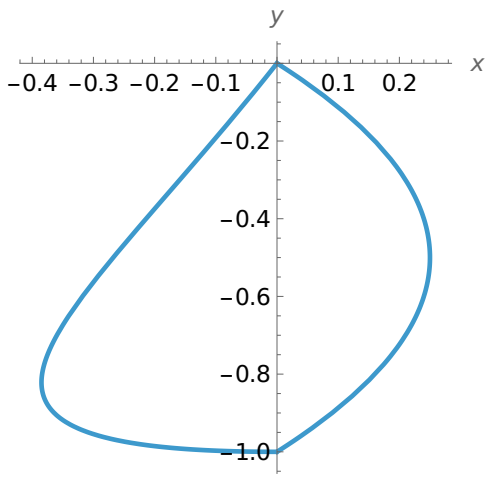
Page 10 of 11 — add any extra pages after page 11 — Page 10 of 11

Question 4 (12 marks)

Consider the curve C in $\mathbb{R}^2$ with parametrisation

$$\mathbf{c}(t) = \begin{cases} (t^3 - t,\, t^2 - 2t), & 0 \le t < 1 \\ (-t^2 + 3t - 2,\, t - 2), & 1 \le t \le 2. \end{cases}$$

It looks like this:



(a) Prove that with the orientation given by increasing t, C is positively oriented.

First observe that $\mathbf{c}(0) = (0, 0) = \mathbf{c}(2)$, and $\mathbf{c}(1) = (0, -1)$.

Next, we have

$$\mathbf{c}'(t) = \begin{cases} (3t^2 - 1, 2t - 2), & 0 \le t < 1 \\ (3 - 2t, 1), & 1 \le t \le 2. \end{cases}$$

In particular the $\mathbf{j}$ component of $\mathbf{c}$ is strictly decreasing for $0 < t < 1$ and strictly increasing for $1 \le t \le 2$. So the first part of C does not cross itself, and the second part of C does not cross itself.

Also note that the $\mathbf{i}$ component of C is negative for $0 < t < 1$ and positive for $1 < t < 2$.

So $\mathbf{c}(t)$ is initially moving downards while it is to the left of the y-axis, and then moving upwards while it is to the right of the y-axis. Hence the curve goes anticlockwise, i.e. it is positively oriented.

Page 11 of 11 — add any extra pages after page 11 — Page 11 of 11

(b) Find the area of the region D inside of C.

We could try to set this up as an integral over D using horizontal or vertical strips, but it's probably easier to use Green's theorem:

$$\iint_D dx\,dy = \frac{1}{2}\int_C x\,dy - y\,dx$$

We must split the integral over C into two parts, but both are just integrals of polynomials. Let C_1 be the part on the left side and C_2 be the part on the right. Then

$$\begin{aligned}\frac{1}{2}\int_{C_1} x\,dy - y\,dx &= \frac{1}{2}\int_0^1 \left(x\frac{dy}{dt} - y\frac{dx}{dt}\right) dt \\ &= \frac{1}{2}\int_0^1 (t^3 - t)(2t-2) - (t^2 - 2t)(3t^2 - 1)\,dt \\ &= \frac{7}{30}\end{aligned}$$

and

$$\begin{aligned}\frac{1}{2}\int_{C_2} x\,dy - y\,dx &= \frac{1}{2}\int_1^2 \left(x\frac{dy}{dt} - y\frac{dx}{dt}\right) dt \\ &= \frac{1}{2}\int_1^2 (-t^2 + 3t - 2)(1) - (t-2)(3-2t)\,dt \\ &= \frac{1}{6}\end{aligned}$$

Adding the two contributions together, we find that the area is

$$\frac{7}{30} + \frac{1}{6} = \frac{2}{5}.$$

End of Assignment — Total Available Marks = 44